CFD Analysis of External Flow Over an Elliptical Aerofoil

1. Objective

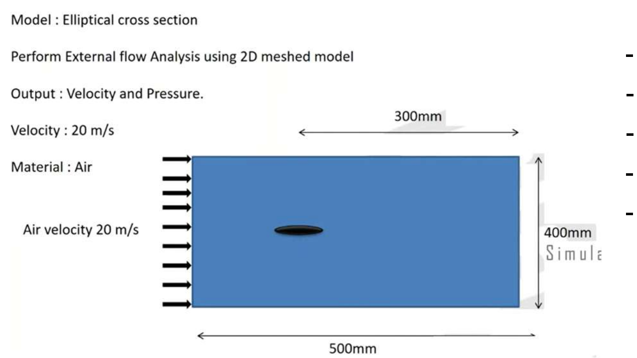
The objective of this project is to perform a 2D external flow analysis over an elliptical aerofoil cross-section using CFD and evaluate the velocity and pressure distribution around the body.

2. Problem Description

A 2D computational domain is created with an elliptical body placed at the center. Air flows from left to right with a uniform inlet velocity.

Given Data:
Flow type: External flow
Fluid: Air
Inlet velocity: 20 m/s
Domain size: 500 mm × 400 mm
Aerofoil type: Elliptical cross-section

- Analysis type: Steady-state

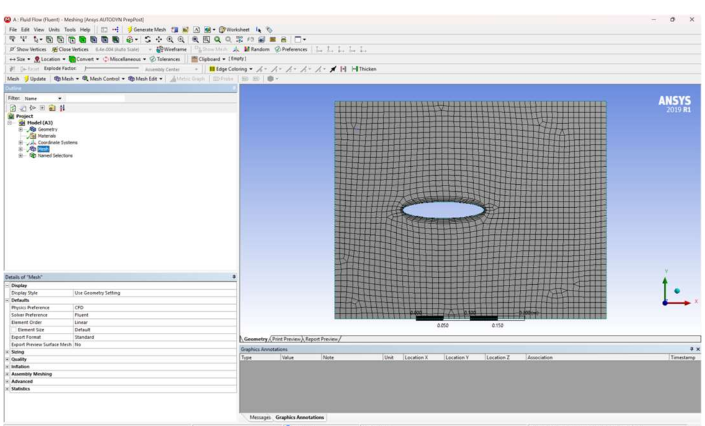


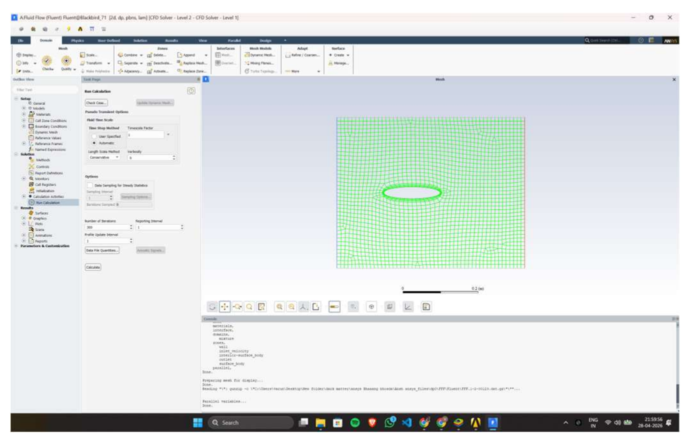
3. Geometry and Domain

The computational domain is rectangular. The upstream length ensures flow development before reaching the aerofoil, while the downstream region captures wake formation. The elliptical body is placed centrally.

4. Meshing

A 2D structured mesh is generated. Mesh refinement is applied near the aerofoil surface to capture boundary layer effects and high velocity gradients. Finer elements are used near the ellipse and coarser mesh away from it.





5. Boundary Conditions

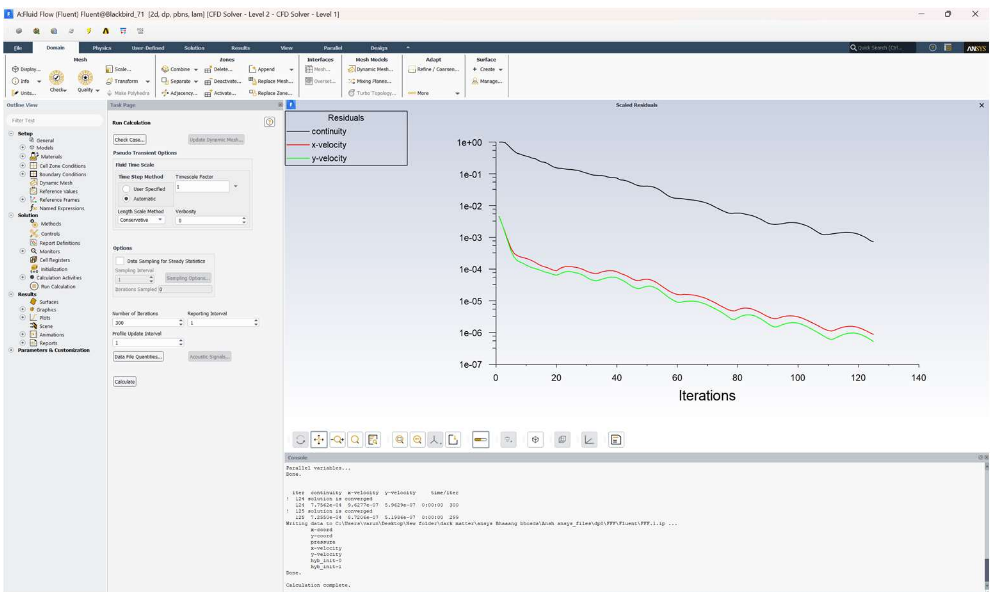
- Inlet: Velocity inlet = 20 m/s
- Outlet: Pressure outlet (atmospheric)
- Aerofoil surface: No-slip wall
- Top & Bottom: Symmetry / far-field

6. Solver Settings

Solver: Pressure-based
Type: Steady-state
Flow model: Laminar
Convergence criteria: Continuity residual ~1e-3, Velocity residual ~1e-6

7. Convergence Behavior

Residuals show stable convergence. Continuity decreases gradually while velocity converges faster. The solution is numerically stable.



8. Results and Discussion

8.1 Pressure Distribution

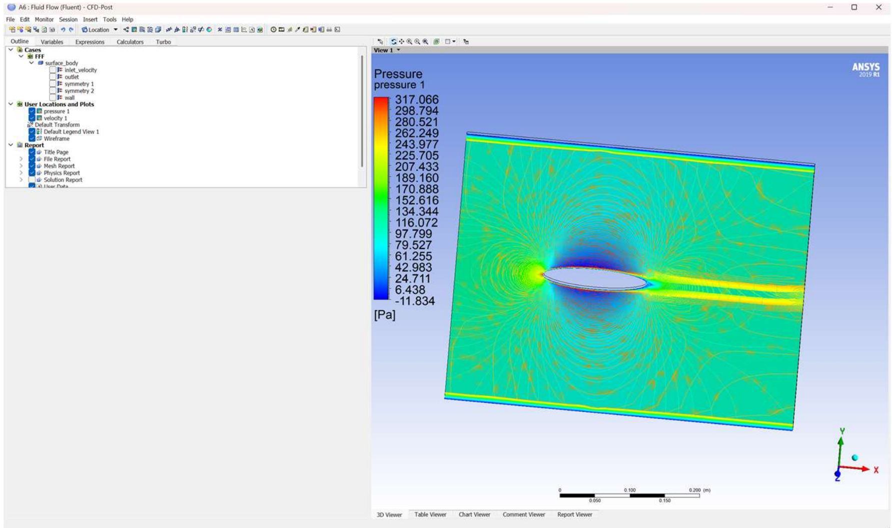
High pressure at the stagnation point, pressure drop along the surface, and recovery in the wake region.

8.2 Velocity Distribution

Acceleration over the aerofoil, maximum velocity at mid-section, and deceleration in the wake.

8.3 Flow Features

Symmetric flow, stagnation point at leading edge, and wake formation behind the aerofoil.



9. Conclusion

The CFD simulation successfully captured the flow behavior around the aerofoil. Results are physically consistent and numerically stable.

10. Future Improvements

Use turbulence models, perform mesh independence study, introduce angle of attack, and compute lift and drag coefficients.